Sean Luka Girgis

Staff Data Engineer | AWS / PySpark Data Platform Engineer

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LinkedIn | GitHub | Portfolio

Professional Summary

Senior Data Engineer with 20+ years of enterprise technology experience designing reliable data platforms, ETL pipelines, data warehouses, telemetry analytics, and performance-focused reporting systems. Strong hands-on background in Python, SQL, PySpark/Spark, AWS S3, Glue, Athena, Redshift, data quality, data modeling, governance, monitoring, and large-scale operational data processing. Experienced in regulated enterprise environments with sensitive data controls, least-privilege thinking, production support, stakeholder communication, and architecture-to-delivery execution. Currently building lakehouse fluency through Databricks, Delta Lake, Unity Catalog, Delta Live Tables, medallion architecture, and Terraform study, grounded in production AWS/PySpark data engineering experience.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Architected and maintained Python/Pandas/PySpark-oriented ETL pipelines ingesting P95 telemetry from 6,000+ enterprise endpoints, replacing manual processes with repeatable, governed data flows for forecasting and executive reporting.
- Designed AWS-enabled data platform patterns using S3 landing zones, AWS Glue ETL, Athena/Redshift access layers, and Oracle-backed historical stores to support large-scale analytics, ML forecasting, and operational decisioning.
- Built and optimized data models, schemas, retention structures, reconciliation checks, and reporting layers for high-volume enterprise infrastructure data, improving trust, traceability, and performance for downstream analytics consumers.
- Integrated disparate monitoring and operational feeds including BMC TrueSight/TSCO, CA Wily/APM, and AppDynamics into unified reporting datasets used by technical teams, capacity planners, and senior stakeholders.
- Developed capacity forecasting workflows using Prophet, scikit-learn, Python, and PySpark concepts to identify bottlenecks months ahead and support proactive infrastructure planning.
- Implemented monitoring, validation, troubleshooting, and runbook-style support around data pipelines and reporting processes, aligning delivery with production reliability and operational readiness expectations.
- Partnered with engineering, infrastructure, and business stakeholders to gather requirements, clarify data definitions, deliver dashboards, and translate complex telemetry into actionable platform and cost-efficiency insights.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Managed Dynatrace AppMon and Gomez Synthetic Monitoring for Brand.com and critical hospitality systems, supporting application performance visibility, transaction tracing, and production monitoring quality.

- Led the FAST project by data-mining real-user and synthetic performance metrics to identify optimization opportunities, creating analytics outputs adopted by engineering teams for customer-facing systems.
- Upgraded Dynatrace from 6.5 to 7.0 and enforced TLS 1.2 monitoring security with no production monitoring regression, balancing modernization, reliability, and security requirements.
- Integrated HP Performance Center with Dynatrace to correlate load testing with APM telemetry, enabling stronger release validation and performance diagnostics.
- Implemented End-User Monitoring and end-to-end transaction tracing across web and mobile experiences, improving operational visibility for business-critical digital channels.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Served as enterprise CA APM SME across 50+ Enterprise Managers and 4,000–6,000 instrumented agents, supporting high-scale monitoring, governance, alerting, dashboards, and production support.
- Led major CA APM upgrades with zero unplanned downtime, coordinating cross-functional teams and maintaining reliability through planning, validation, communication, and technical documentation.
- Designed custom management modules, dashboards, SLA thresholds, and alerting standards that improved operational consistency and stakeholder visibility across client environments.
- Built Perl and Korn Shell extraction scripts to automate performance reporting and replace manual weekly exports, improving repeatability and delivery speed.
- Provided sizing guidance, onboarding patterns, technical training, and documentation to help teams adopt monitoring standards across complex enterprise applications.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster while preserving transactional performance and improving platform manageability.
- Optimized C++/OCCI transaction processing and SQL/database access patterns, reducing hardware footprint while maintaining low-latency query behavior at enterprise scale.
- Built C++/OCCI/OCI validation tooling to verify migration correctness across large transaction datasets and support phased production cutover.
- Automated schema conversion, reconciliation, and data validation tasks to reduce migration risk and improve confidence in platform modernization.

Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (AMDOCS Projects) (2001 – 2007)

- Developed high-availability C/C++/Pro*C/PL/SQL billing interfaces and reporting processes across telecom billing systems, supporting large-scale data movement, integration, and production operations.
- Delivered performance improvements in billing and database processes, including memory reduction, throughput gains, and SQL/sequence optimization for high-volume transactional workloads.
- Built and maintained XML, MQ Series, WebLogic/WebSphere, Korn Shell, Perl, Oracle, and UNIX-based integration components across full billing lifecycle workflows.
- Reverse-engineered and documented undocumented vendor APIs to enable interface integration, production support, and knowledge transfer across engineering teams.

Education

High Diploma / Post-Graduate Diploma in Computer Engineering Technology / Computer Science
Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering
Zagazig University, Egypt / Cairo, Egypt

Skills

Languages: Python, SQL, PySpark, PL/SQL, Perl, Korn Shell, C, C++, Java, XML, JSON

Flagship Projects

Serverless Lakehouse Platform (AWS)
Designed a serverless data platform pattern using S3, Glue, Athena, Redshift, PySpark, Parquet, and Bedrock concepts to support governed analytics, optimized ETL, and AI-assisted data access.
The project demonstrates lakehouse thinking, small-file mitigation, columnar storage, and enterprise-scale data access patterns relevant to Databricks and medallion architecture roles.
Technologies: AWS S3, AWS Glue, Athena, Redshift, PySpark, Parquet, Bedrock, Data Lakehouse

HorizonScale — AI Capacity Forecasting Engine
Built an AI-driven capacity forecasting engine using Python, PySpark, Prophet, scikit-learn, and Streamlit to transform telemetry data into forward-looking infrastructure risk insights.
The project maps operational telemetry into data products, dashboards, and forecasting outputs for stakeholder decision-making.
Technologies: Python, PySpark, Prophet, scikit-learn, Streamlit, ETL, Telemetry Analytics, Forecasting

AI-Powered Job Search Pipeline
Built an agentic data and document workflow that performs semantic duplicate detection, scoring, JSON artifact generation, quality gating, and application tracking.
The system uses structured contracts, vector embeddings, local automation, and LLM APIs to turn unstructured postings into governed downstream artifacts.
Technologies: Python, FAISS, Sentence Transformers, LLM APIs, JSON Contracts, Quality Gates, Automation Pipelines, RAG Concepts

FAST Project
Data-mined Dynatrace and Gomez Synthetic Monitoring performance metrics to identify optimization opportunities for customer-facing systems, producing analytics outputs that engineering teams used to improve performance and reliability.
Technologies: Dynatrace, Gomez Synthetics, Data Mining, Performance Analytics, Operational Reporting